EXPLORING BUSINESS · CHAPTER 7

From Product Design and Development...

Revision sheet

IN ONE SENTENCE

A product provides a **benefit** and satisfies a **need**. Before developing it, you check that the idea is an **opportunity** (commercial potential), narrow your analysis from the **industry** down to the **niche**, **forecast demand**, then follow the **7 steps** of product development and protect the idea with a **patent**.

FOUR TYPES OF NEW PRODUCTS

Type	Example
New to the company	Peeps decorating kits
Improvement	Mike & Ike's in pouches (+20%)
Line extension	Peeps in a chocolate egg
New to the market (≈ 10%)	PowerSki Jetboard

ENTREPRENEURIAL START-UP

- **Innovative** products or practices.
- Goals of **profitability and growth**.
- Focus on **new opportunities**.
- Owners willing to take **calculated risks** (research + experience).

IDEAS

Right question

"What does the customer want to buy?"

Purple cow

A truly remarkable product (Seth Godin).

Failures

≈ 30,000 launches a year, up to 90% fail.

Sources

Internal (marketing, sales, R&D, manufacturing) and external (customers, competitors).

FOUR UTILITIES

Utility	Example
Time	Water at a concert
Place	Hot dogs outside offices
Ownership	Real estate agent
Form	Fabric → apparel

FROM INDUSTRY TO NICHE

Industry (competing businesses) → Market (buyers, one need) →

Segment (common characteristics) → Niche (specific need)

SEGMENTATION

Demographic

age, sex, income, education, ethnicity, occupation, life stage

Geographic

region, climate, city size, urban/rural

Psychographic

lifestyle, values, personality, attitude to risk

FORECASTING DEMAND

- Talk with **similar businesses**.
- Talk with **potential customers**.
- Examine **published industry data**.
- **Market share**: 600,000 × 2% = 12,000 pairs.

BREAKEVEN (NOT FOR THE MCQ)

Formula	`fixed costs ÷ (price − variable cost)`
Example	`300,000 ÷ (80 − 45) = 8,571 pairs`
Grooming at $35	`164,000 ÷ 28.85 → 5,685 dogs`
Burgers at $2.10	`180,000 ÷ 1.15 → 156,522`

THE 7 DEVELOPMENT STEPS

Select the idea → Refine the concept →

Performance and appeal →

Design for manufacturing → Prototypes →

Ramp-up and tests → Launch

DEVELOPMENT: RISKS

- **12 failures** for every success; $60bn wasted a year (Fortune 1000).
- Three causes: **trade-offs**, **time pressure**, **economics**.
- A cross-functional **project team**.

ARTIFICIAL INTELLIGENCE

AI

Tasks commonly associated with intelligent beings.

Machine learning

Learning from data; neural networks.

Training / test set

Labeled pictures, then new pictures.

Jobs

Cat. 1 automated (to fear); cat. 2 made more effective (to admire).

PATENT

Granted by

USPTO

Right

To exclude others from making, using, selling

Duration

20 years, one country only

Criteria

New, not obvious, has utility

TRAPS

- Industry = **businesses**; market = **buyers**.
- The **concept** is a description; the **prototype** is a physical model.
- A U.S. patent does not protect abroad.
- Breakeven analysis is **not for the MCQ**.